

One Fold if and only if the Overlap Law is Right-Skewed: A Riccati Criterion for Tilted Beta Normalizers, with Consequences for Rank-One Projective Rate–Distortion

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Abstract

For $M = {}_1F_1(a; c; \kappa)$, the moment generating function of $X \sim \text{Beta}(a, c - a)$, the stationary multiplier $\lambda(\kappa) = \kappa/(2b(\kappa))$ of the Curie–Weiss-type radial problem $\max_{0 \leq b \leq 1} \{K(2\lambda b) - \lambda R^2 b^2\}$, $K = \log M - (a/c)\kappa$, folds where $H = K' - \kappa K''$ vanishes. We show that the logarithmic derivative $m = M'/M$ satisfies the Riccati equation $\kappa m' = a - (c - \kappa)m - \kappa m^2$, so that $H = m(1 - m)(\Psi(m) - \kappa)$ with Ψ an explicit rational function, and a transversality argument in the plane (m, κ) proves: H has exactly one positive zero if $c > 2a$ (right-skewed overlap law) and none if $c \leq 2a$. This settles the second open question of ARR-2026-68BH5JJCQ8XW8GT, gives the complete scalar radial phase and two-piece envelope for every (a, c) with explicit fold bounds for $a \geq 1$ (false for $a < 1$), covers the Bessel family ${}_0F_1(; c; \kappa^2/4)$ (no fold), and yields “one fold iff $d \geq 3$ ” for rank-one sources over $\mathbb{R}, \mathbb{C}, \mathbb{H}$. For the oriented-two-plane normalizer, which is third order, we give an exact identity and numerics only.

1. Introduction

Antecedents. The record ARR-2026-68BH5JJCQ8XW8GT (*One Fold, Unique Contact and the Complete Two-Piece Rate–Distortion Function of Rank-One Complex-Projective Born Prediction*; below “68BH”) proved that for $M_d = {}_1F_1(1; d; \kappa)$, $d \geq 3$, the fold numerator $H_d = K'_d - \kappa K''_d$ has exactly one positive zero (its Theorem 4.1), and derived from this, through the standing facts (H1)–(H6) of its Section 5, the complete radial phase and the two-piece rate–distortion function of Haar rank-one Born prediction on $\mathbb{CP}^{d-1}$ (its Theorem 5.1, closing Remark 3.2 of ARR-2026-1D2QYXPCVY9H7ANB, “1D2Q”, whose Theorem 2.1 is the envelope formula). The tool there was the inhomogeneous first-order equation $\kappa M'_d = (\kappa - d + 1)M_d + (d - 1)$ (68BH Lemma 3.1), available because the first Kummer parameter equals 1; 68BH’s second open question asked for a criterion in (a, c) for ${}_1F_1(a; c; \kappa)$, anticipating that “the (M, κ) transversality argument must be replaced by one in a three-dimensional phase space”. Sibling records reached one-fold statements in other geometries by coefficient sign patterns: ARR-2026-61Y0FFA39M8KMBJ5 (“61Y0”, Lemma 6.1, Theorem 6.2, Corollary 6.3) for $\text{Gr}_{\mathbb{C}}(2, 4)$, and ARR-2026-7H9FAPTBA897AMJ (“7H9F”, Lemma 7.1, Theorems 7.2, 8.2, 9.1) for oriented two-planes in $\mathbb{R}^n$, whose normalizer $L_q = \mathbb{E} \cosh(\kappa T)$, $T \sim \text{Beta}(1, q)$, $q = n - 2$, is the even part of M_{q+1} . Phase-lifted coherent orbits (ARR-2026-6XH6JAS5ZA934A6J, “6XH6”; ARR-2026-6DJ302B1G38V4SHD, “6DJ3”) have ${}_pF_{p+1}$ normalizers, the simplest being the Bessel function ${}_0F_1(; n; \kappa^2/4)$ (6XH6 Proposition 7.1, $k = 1$). Externally: Kummer’s equation, its integral representation and its large-argument expansion are DLMF §13.2, §13.4, §13.7 [DLMF13]; Kummer’s second theorem, relating ${}_1F_1(\nu + \frac{1}{2}; 2\nu + 1; 2z)$ to $I_\nu(z)$,

is DLMF 13.6.9; the reduction of a second-order linear equation to a first-order Riccati equation for the logarithmic derivative is classical [Ince]; the uniform law on S^{n-1} has a smooth rate-distortion curve (Dytso–Cardone [DC]); generalized Curie–Weiss models may have several first-order transitions (Eisele–Ellis [EE]); the real and complex Bingham normalizers are ${}_1F_1(\frac{1}{2}; \frac{d}{2}; \cdot)$ and ${}_1F_1(1; d; \cdot)$ (Bingham [B], Kent [K]); and the Kummer-ratio literature (Karp–Sitnik [KS], Kalmykov–Karp [KK]) concerns monotonicity of ratios such as $m = M'/M$, not the sign of $K' - \kappa K''$.

Contribution. (i) A one-parameter Riccati reduction (Theorem 3.1): for every $a > 0, c > a$, the state needed to decide the sign of H is (m, κ) , two-dimensional, not three. (ii) The criterion (Theorem 4.1): H has exactly one positive zero if $c > 2a$ and none if $c \leq 2a$; equivalently, one fold iff the overlap law $\text{Beta}(a, c - a)$ is right-skewed. (iii) Explicit bounds on the fold for $a \geq 1$ (Theorem 5.1), with counterexamples for $a < 1$. (iv) The complete scalar radial phase and its two-piece envelope for every (a, c) (Theorem 6.1), with a precise list of what must be imported to read the envelope as a rate-distortion function. (v) No fold for the Bessel family (Theorem 7.1, Proposition 7.2) and “one fold iff $d \geq 3$ ” for rank-one sources over the three fields (Theorem 7.4). (vi) For the third-order oriented-two-plane normalizer, an exact identity $(\kappa^{2q+2} L_q^3 H_q)' = \kappa^{2q+2} L_q^3 \Delta_q$ and exact small-field signs (Section 8), but no new proof of 7H9F Theorem 8.2. 68BH’s conventions are restated in Section 2 and every step of the chain to the envelope is written out. Findings of an independent adversarial review are incorporated (Section 7: the range $c \geq 1/2$; Section 8: Proposition 8.2).

2. Setting

Throughout $a > 0, c > a, \kappa > 0$, and $X \sim \text{Beta}(a, c - a)$ with density $x^{a-1}(1-x)^{c-a-1}/B(a, c-a)$ on $(0, 1)$. By DLMF 13.4.1,

$$M(\kappa) := \mathbb{E} e^{\kappa X} = {}_1F_1(a; c; \kappa) = \sum_{j \geq 0} \frac{(a)_j}{(c)_j} \frac{\kappa^j}{j!}, \quad \mu := \mathbb{E} X = \frac{a}{c}, \quad R^2 := 1 - \mu. \quad (2.1)$$

M is entire, positive and strictly increasing on $[0, \infty)$. Put $K(\kappa) = \log M(\kappa) - \mu\kappa$ (the cumulant generating function of $X - \mu$), $m = M'/M = \mathbb{E}_\kappa X$ (the tilted mean under the law $\propto e^{\kappa x}$), so $K' = m - \mu$ and $K'' = m' = \text{Var}_\kappa X > 0$. The radial problem and its stationary parametrization are

$$G_\lambda(b) = K(2\lambda b) - \lambda R^2 b^2 \quad (0 \leq b \leq 1), \quad b(\kappa) = \frac{K'(\kappa)}{R^2} = \frac{m(\kappa) - \mu}{1 - \mu}, \quad \lambda(\kappa) = \frac{\kappa}{2b(\kappa)}, \quad (2.2)$$

and the fold numerator and coexistence function are

$$H := K' - \kappa K'' = (m - \mu) - \kappa m', \quad F := 2K - \kappa K'. \quad (2.3)$$

Direct differentiation gives $F' = H$, $\lambda' = H/(2R^2 b^2)$, and at a stationary radius ($K'(\kappa) = R^2 b$, $\lambda = \kappa/(2b)$) $G_\lambda(b) = F(\kappa)/2$ and $\partial_b^2 G_\lambda(b) = -(2\lambda/b)H(\kappa)$ (manuscript_checks.py). For $a = 1, c = d$ these are (2.1)–(2.4) of 68BH: $R^2 = R_{0,d}^2 = 1 - 1/d$, $K = K_d$, $H = H_d$, $F = F_d$, $\lambda_0 = d(d+1)/2$. The constant R^2 plays no role in the zero counts of H and F ; it fixes b and λ .

Sources with this overlap law. For x Haar on the unit sphere of $\mathbb{F}^d$ ($\mathbb{F} = \mathbb{R}, \mathbb{C}, \mathbb{H}$, $\beta = 1, 2, 4$) and a fixed unit u , $|\langle x, u \rangle|^2 \sim \text{Beta}(\beta/2, \beta(d-1)/2)$, giving $(a, c) = (\frac{1}{2}, \frac{d}{2})$, $(1, d)$, $(2, 2d)$: the real Bingham/Watson, complex Bingham and quaternionic normalizers $\mathbb{E} e^{\kappa |\langle x, u \rangle|^2}$. The case $\beta = 2$ is the source of 1D2Q/68BH. Two symmetric families are treated in Sections 7–8: $Z_c(\kappa) = {}_0F_1(; c; \kappa^2/4)$ (spheres) and L_q (oriented two-planes).

3. The Riccati reduction

Theorem 3.1. For every $a > 0$, $c > a$ and $\kappa > 0$,

$$\kappa m' = a - (c - \kappa)m - \kappa m^2, \quad (3.1)$$

$$H = -\kappa m(1 - m) + (c + 1)(m - \mu) = m(1 - m)(\Psi(m) - \kappa), \quad \Psi(m) := \frac{(c + 1)(m - \mu)}{m(1 - m)}. \quad (3.2)$$

In particular $\text{sign } H(\kappa) = \text{sign}(\Psi(m(\kappa)) - \kappa)$.

Proof. Kummer's equation $\kappa M'' + (c - \kappa)M' - aM = 0$ (DLMF 13.2.1) divided by M , with $M''/M = m' + m^2$, is (3.1). Then $H = (m - \mu) - \kappa m' = (m - \mu) - a + (c - \kappa)m + \kappa m^2 = -\kappa m(1 - m) + (c + 1)m - (c + 1)\mu$, using $a = c\mu$. The factorization is immediate. $\square$

This is the general- (a, c) analogue of 68BH Lemma 3.2. No inhomogeneous first-order equation is needed: the logarithmic derivative of any solution of a second-order linear equation satisfies a first-order Riccati equation [Ince]. The relevant state is therefore (m, κ) , and 68BH's expectation of a three-dimensional phase space for $a \neq 1$ was too pessimistic; the third dimension appears only for higher-order normalizers (Section 8).

Lemma 3.2 (behaviour at the ends). (i) With κ_j the cumulants of X , $H = \sum_{j \geq 3} \kappa_j(2 - j)\kappa^{j-1}/(j - 1)!$; hence

$$H(\kappa) = -\frac{\mu_3}{2}\kappa^2 - \frac{\kappa_4}{3}\kappa^3 + O(\kappa^4), \quad \mu_3 = \frac{2a(c - a)(c - 2a)}{c^3(c + 1)(c + 2)}, \quad (3.3)$$

and at $c = 2a$ (where $\mu_3 = 0$) the cubic coefficient is $-\kappa_4/3 = 1/(8(2a + 1)^2(2a + 3)) > 0$. (ii) $m(\kappa) = 1 - (c - a)/\kappa + O(\kappa^{-2})$ and $\kappa m' \rightarrow 0$; hence $H \rightarrow 1 - \mu > 0$ as $\kappa \rightarrow \infty$. (iii) $\log M(\kappa) = \kappa - (c - a) \log \kappa + \log(\Gamma(c)/\Gamma(a)) + o(1)$, so $F(\kappa) = R^2\kappa - 2(c - a) \log \kappa + O(1) \rightarrow +\infty$. (iv) $m : (0, \infty) \rightarrow (\mu, 1)$ is a C^∞ increasing bijection.

Proof. (i) $K = \sum_{j \geq 2} \kappa_j \kappa^j / j!$ termwise; the third central moment of $\text{Beta}(a, c - a)$ from $\mathbb{E}X^r = (a)_r / (c)_r$ and κ_4 at $c = 2a$ are the displayed values (manuscript_checks.py). (ii) DLMF 13.7.1 gives ${}_1F_1(\alpha; \gamma; \kappa) = e^\kappa \kappa^{\alpha - \gamma} \frac{\Gamma(\gamma)}{\Gamma(\alpha)} (1 + \frac{(1 - \alpha)(\gamma - \alpha)}{\kappa} + O(\kappa^{-2}))$; applying it to $m = \frac{a}{c} {}_1F_1(a + 1; c + 1; \kappa) / {}_1F_1(a; c; \kappa)$ gives the expansion of m , and (3.1) then gives $\kappa m' = a - cm + \kappa m(1 - m) = O(\kappa^{-1})$. (iii) is the leading term of the same expansion; $F = 2 \log M - \mu \kappa - \kappa m$. (iv) $m' = \text{Var}_\kappa X > 0$, $m(0^+) = \mu$, and $m \rightarrow 1$ by (ii). $\square$

4. The one-fold criterion

Theorem 4.1. Let $a > 0$, $c > a$.

(A) If $c > 2a$, then H has exactly one positive zero κ_f , with $H < 0$ on $(0, \kappa_f)$ and $H > 0$ on (κ_f, ∞) ; moreover $m_f := m(\kappa_f) \geq \frac{1}{2}$ and

$$\kappa_f = \Psi(m_f) \geq \Psi\left(\frac{1}{2}\right) = \frac{2(c + 1)(c - 2a)}{c}. \quad (4.1)$$

Consequently $\lambda(\kappa)$ decreases strictly on $(0, \kappa_f]$ from $\lambda_0 := \lambda(0^+) = c(c + 1)/(2a)$ to $\lambda_{\min} := \lambda(\kappa_f)$ and increases strictly on $[\kappa_f, \infty)$ to $+\infty$.

(B) If $c \leq 2a$, then $H > 0$ on $(0, \infty)$, and λ increases strictly from λ_0 to $+\infty$.

Proof. By Lemma 3.2(iv) let $\kappa(m)$ be the inverse of $m(\kappa)$ on $(\mu, 1)$ and set $D(m) := \Psi(m) - \kappa(m)$, so that $\text{sign } H(\kappa) = \text{sign } D(m(\kappa))$ by (3.2). By (3.1), $d\kappa/dm = 1/m'(\kappa) = G(m, \kappa(m))$ with

$$G(m, \kappa) := \frac{\kappa}{a - cm + \kappa m(1 - m)}, \quad D'(m) = \Psi'(m) - G(m, \kappa(m)).$$

The following are rational identities in (a, c, m) (`manuscript_checks.py`):

- (a) $\Psi'(m) = (c + 1)[(m - \mu)^2 + \mu(1 - \mu)]/(m(1 - m))^2 > 0$ on $(\mu, 1)$.
- (b) On the curve $\kappa = \Psi(m)$ the denominator of G equals $m - \mu > 0$, and $G(m, \Psi(m)) = \Psi(m)/(m - \mu)$.
- (c) $\Delta(m) := \Psi'(m) - G(m, \Psi(m)) = -(c + 1)(m - \mu)(1 - 2m)/(m(1 - m))^2$; on $(\mu, 1)$ it has the sign of $2m - 1$.
- (d) $\partial_\kappa G(m, \kappa) = (a - cm)/(a - cm + \kappa m(1 - m))^2 < 0$ for $m > \mu$ wherever the denominator is nonzero; the denominator is affine and increasing in κ .

At a zero m_0 of D one has $\kappa(m_0) = \Psi(m_0)$, hence $D'(m_0) = \Delta(m_0)$. From Lemma 3.2: (e) $D < 0$ near μ^+ iff $H < 0$ near 0^+ iff $\mu_3 > 0$ iff $c > 2a$, while for $c \leq 2a$ $D > 0$ near μ^+ ($\mu_3 < 0$, or the positive cubic coefficient at $c = 2a$); (f) $D > 0$ near 1^- since $H \rightarrow 1 - \mu > 0$.

Case $c > 2a$, i.e. $\mu < \frac{1}{2}$. Step 1 ($D < 0$ on $(\mu, \frac{1}{2})$, $D(\frac{1}{2}) \leq 0$): if $D \geq 0$ somewhere in $(\mu, \frac{1}{2})$, let m_* be the infimum of such points; by (e) $m_* > \mu$, $D(m_*) = 0$ and $D < 0$ before, so $D'(m_*) \geq 0$; but $D'(m_*) = \Delta(m_*) < 0$ by (c). Step 2 (on $[\frac{1}{2}, 1)$, $D < 0 \Rightarrow D' > 0$): if $D(m) < 0$ then $\kappa(m) > \Psi(m)$; the denominator of $G(m, \cdot)$ is positive at $\kappa(m)$ (it equals $\kappa m' > 0$) and at $\Psi(m)$ (by (b)), hence on the whole segment $[\Psi(m), \kappa(m)]$, where (d) gives $G(m, \kappa(m)) < G(m, \Psi(m))$; so $D'(m) > \Delta(m) \geq 0$, strictly even at $m = \frac{1}{2}$. Step 3: let $m_f := \inf\{m \geq \frac{1}{2} : D(m) \geq 0\}$, which exists by (f); $D(m_f) = 0$ and $D < 0$ on (μ, m_f) by Step 1. If $D(m_2) < 0$ for some $m_2 > m_f$, let $m_3 = \sup\{m \in [m_f, m_2] : D(m) \geq 0\}$; then $D(m_3) = 0$, $D < 0$ on $(m_3, m_2]$, where Step 2 makes D strictly increasing, so $D(m_2) > 0$: contradiction. Thus $D \geq 0$ on $[m_f, \infty)$, and a zero $m_0 > m_f \geq \frac{1}{2}$ would be a local minimum with $D'(m_0) = \Delta(m_0) > 0$: impossible. Hence $D < 0$ on (μ, m_f) , $D > 0$ on $(m_f, 1)$, i.e. H has the unique zero $\kappa_f = \kappa(m_f)$ with the stated signs, $m_f \geq \frac{1}{2}$, and $\kappa_f = \Psi(m_f) \geq \Psi(\frac{1}{2})$ because Ψ is increasing by (a); $\Psi(\frac{1}{2})$ is a direct evaluation. (A tangential zero at $m_f = \frac{1}{2}$, where $\Delta = 0$, is not excluded and does not affect the argument.)

Case $c \leq 2a$, i.e. $\mu \geq \frac{1}{2}$. Now $\Delta > 0$ on $(\mu, 1)$ by (c) and $D > 0$ near μ^+ by (e). A first zero m_* of D would satisfy $D'(m_*) \leq 0$, contradicting $D'(m_*) = \Delta(m_*) > 0$. So $D > 0$, i.e. $H > 0$, on $(0, \infty)$.

The statements on λ follow from $\lambda' = H/(2R^2b^2)$, $\lambda(0^+) = R^2/(2K''(0)) = R^2/(2v)$ with $v = \text{Var}X = a(c - a)/(c^2(c + 1))$, which equals $c(c + 1)/(2a)$, and $\lambda(\kappa) = \kappa/(2b) > \kappa/2 \rightarrow \infty$ since $b < 1$. $\square$

Remarks. (1) For $a = 1, c = d$, Theorem 4.1 reproduces 68BH Theorem 4.1 and Corollary 4.2, including the lower bound $2(d - 2)(d + 1)/d$; the phase plane differs $((M, \kappa)$ there, (m, κ) here), the argument is the same. (2) The threshold is forced by two facts that do not depend on the fine structure of M : the transversality sign Δ changes at $m = \frac{1}{2}$ for every (a, c) , and a fold requires the trajectory to start below $\frac{1}{2}$ ($\mu < \frac{1}{2}$) with $H < 0$ there ($\mu_3 > 0$); both say $c > 2a$. (3) At the fold the tilted mean is at least $\frac{1}{2}$, i.e. $b_f := b(\kappa_f) \geq (c - 2a)/(2(c - a))$.

5. Bounds on the fold for $a \geq 1$, and their failure for $a < 1$

Theorem 5.1. Let $c > 2a$ and $a \geq 1$. Then $\kappa(1 - m(\kappa)) < c - a$ for all $\kappa > 0$, and

$$\frac{1}{2} \leq m_f < m_u := \frac{a(c+1)}{c(a+1)}, \quad \frac{2(c+1)(c-2a)}{c} \leq \kappa_f < \Psi(m_u) = c(a+1). \quad (5.1)$$

Proof. Write $Z(\kappa) = \int_0^1 x^{a-1}(1-x)^{c-a-1}e^{\kappa x}dx = B(a, c-a)M(\kappa)$ and integrate by parts against $d[(1-x)^{c-a}]$: $(c-a)Z = -\int_0^1 x^{a-1}e^{\kappa x}d[(1-x)^{c-a}] = [x^{a-1}e^{\kappa x}(1-x)^{c-a}]_1^0 + \int_0^1 (1-x)^{c-a}((a-1)x^{a-2} + \kappa x^{a-1})e^{\kappa x}dx$. The boundary term is $[a=1]$ (it vanishes at $x=1$ and at $x=0$ for $a > 1$), and for $a \geq 1$ the remaining integrand is integrable (x^{a-2} is integrable at 0 iff $a > 1$, and the term is absent for $a=1$). Dividing by Z ,

$$c-a = \frac{[a=1]}{Z(\kappa)} + (a-1)\mathbb{E}_\kappa\left[\frac{1-X}{X}\right] + \kappa\mathbb{E}_\kappa[1-X] \quad (a \geq 1). \quad (5.2)$$

All three terms are ≥ 0 and the last is > 0 , so $\kappa(1-m) < c-a$. Now (3.2) can be rearranged as $H = m[(c-a) - \kappa(1-m)] + (a+1)m - (c+1)\mu$, hence $H > (a+1)m - (c+1)\mu \geq 0$ whenever $m \geq m_u$. Therefore $m_f < m_u$; since $\mu < m_u < 1$ (as $a < c$) and Ψ is increasing, $\kappa_f = \Psi(m_f) < \Psi(m_u)$, and $\Psi(m_u) = c(a+1)$ is a rational identity (manuscript_checks.py). $\square$

For $a=1$, $c=d$, (5.1) is 68BH's $\kappa_{f,d} < 2d$ and its Remark 3.3 ($b_f < \frac{1}{2}$, equivalent to $m_f < (d+1)/(2d)$).

Proposition 5.2 (failure for $a < 1$). For $a < 1$ the identity (5.2) does not hold (the term x^{a-2} is not integrable at 0), the inequality $\kappa(1-m) \leq c-a$ fails, and so does $\kappa_f < c(a+1)$. Numerically (checks_bounds.py, tables_ac.py; independently reproduced in review): $\sup[\kappa(1-m) - (c-a)]$ over the tested grid is 2.46 for $a = \frac{1}{4}$, 1.15 for $a = \frac{1}{2}$, 0.41 for $a = \frac{3}{4}$, and $\leq 10^{-27}$ for $a=1$; and $(a, c) = (\frac{1}{2}, 2)$ has $\kappa_f = 3.8790 > 3 = c(a+1)$, $(1/4, 1)$ has $\kappa_f = 2.8009 > 1.25$ (Table 4). No upper bound on κ_f is claimed for $a < 1$; the lower bounds of (4.1) hold for every a .

6. Consequences for the rate-distortion chain

Standing facts, valid for all $a > 0$, $c > a$, $\kappa > 0$, $\lambda > 0$ (the analogues of 68BH (H1)–(H6)):

(H1) $b' = m'/R^2 > 0$, $b(0^+) = 0$, $b(\kappa) \rightarrow 1$ (Lemma 3.2(iv)).

(H2) $K'(\kappa) = m - \mu < 1 - \mu = R^2$ because $X < 1$ a.s.; hence $G'_\lambda(1) = 2\lambda(K'(2\lambda) - R^2) < 0$: $b=1$ is never stationary and never a maximizer.

(H3) For $b \in (0, 1]$: $G'_\lambda(b) = 0$ iff $b = b(\kappa)$ and $\lambda = \lambda(\kappa)$ with $\kappa = 2\lambda b$; $\kappa \mapsto b(\kappa)$ is injective by (H1).

(H4) At such a radius $G_\lambda(b) = F(\kappa)/2$ and $G''_\lambda(b) = -(2\lambda/b)H(\kappa)$ (Section 2).

(H5) $F' = H$, $F(0) = 0$, $F \rightarrow +\infty$ (Lemma 3.2(iii)).

(H6) $G_\lambda(b) = \lambda(2\lambda v - R^2)b^2 + \frac{4}{3}\lambda^3\mu_3b^3 + O(b^4)$ with $v = a(c-a)/(c^2(c+1))$, $\lambda_0 = R^2/(2v) = c(c+1)/(2a)$; so 0 is a strict local maximum of G_λ for $\lambda < \lambda_0$, a strict local minimum for $\lambda > \lambda_0$, and for $c > 2a$ (where $\mu_3 > 0$) not a local maximum at $\lambda = \lambda_0$.

Theorem 6.1 (complete scalar radial phase and envelope). Let $a > 0$, $c > a$, and define $\tilde{c}(\lambda) := \max_{0 \leq b \leq 1} G_\lambda(b)$ and, for $0 < D \leq R^2$, the envelope

$$\mathcal{R}(D) := \sup_{\lambda \geq 0} \{\lambda(R^2 - D) - \tilde{c}(\lambda)\}. \quad (6.1)$$

(I) *Case $c > 2a$.* Let $\kappa_f, \lambda_{\min}$ be as in Theorem 4.1(A). (a) The positive stationary radii of G_λ are the $b(\kappa)$ with $\lambda(\kappa) = \lambda$: none for $\lambda < \lambda_{\min}$; one ($b(\kappa_f)$, degenerate, with gain $F(\kappa_f)/2 < 0$) at $\lambda_{\min}$; two, $b(\kappa_-) < b(\kappa_+)$ with $\kappa_- < \kappa_f < \kappa_+$, for $\lambda_{\min} < \lambda < \lambda_0$; one, $b(\kappa_+)$ with $\kappa_+ > \kappa_f$, for $\lambda \geq \lambda_0$. Pre-fold radii are strict local minima, post-fold radii strict local maxima. (b) F decreases strictly on $(0, \kappa_f]$, is negative there, then increases strictly to $+\infty$: it has a unique positive zero $\kappa_c > \kappa_f$. Put $b_c = b(\kappa_c)$, $\lambda_c = \lambda(\kappa_c)$, $D_c = R^2(1 - b_c^2)$, $R_c = \kappa_c K'(\kappa_c) - K(\kappa_c) = \lambda_c R^2 b_c^2$. (c) The global maximizers of G_λ on $[0, 1]$ are $\{0\}$ for $0 < \lambda < \lambda_c$, $\{0, b_c\}$ at λ_c , $\{b(\kappa_+(\lambda))\}$ for $\lambda > \lambda_c$, where $\kappa_+(\lambda) > \kappa_c$ is the unique post-fold solution of $\lambda(\kappa) = \lambda$. (d) $0 < \lambda_{\min} < \lambda_c < \lambda_0$. (e) On (λ_c, ∞) , $\lambda \mapsto b(\kappa_+(\lambda))$ is continuous and strictly increasing to 1 (no reentrance); $\tilde{c}$ is convex with $-\partial\tilde{c}(\lambda) = \{R^2\}$ for $\lambda < \lambda_c$, $[D_c, R^2]$ at λ_c , $\{D(\lambda)\}$ for $\lambda > \lambda_c$, $D(\lambda) := R^2(1 - b(\kappa_+(\lambda))^2)$. (f) $\mathcal{R}(D) = \lambda_c(R^2 - D)$ for $D_c \leq D \leq R^2$, and $\mathcal{R}(D) = \kappa K'(\kappa) - K(\kappa)$ for $0 < D < D_c$, where $\kappa > \kappa_c$ is the unique solution of $b(\kappa) = \sqrt{1 - D/R^2}$; $\mathcal{R}(D) \rightarrow \infty$ as $D \downarrow 0$.

(II) *Case $c \leq 2a$.* For $0 < \lambda \leq \lambda_0$ the unique global maximizer is 0; for $\lambda > \lambda_0$ it is the unique positive stationary radius $b_+(\lambda)$, which increases continuously from 0 (as $\lambda \downarrow \lambda_0$) to 1: continuous onset, no coexistence, no reentrance. $\mathcal{R}(D) = \kappa K'(\kappa) - K(\kappa)$ for $0 < D < R^2$ with $b(\kappa) = \sqrt{1 - D/R^2}$.

Proof. (I)(a)–(e) are word for word the proof of 68BH Theorem 5.1(a)–(e), which uses only the sign pattern of H (Theorem 4.1(A)), $F' = H$ and (H1)–(H6); we indicate the steps. (a) is (H3) with the counts from the monotonicity of $\lambda(\kappa)$ (the value λ_0 is a limit, not attained on $(0, \kappa_f]$), the local type from (H4), and the degenerate gain from (b). (b) is $F' = H$, $F(0) = 0$, (H5). (c): a maximizer exists among $\{0\}$, $\{1\}$ and the interior stationary points; 1 is excluded by (H2), pre-fold radii and the degenerate radius by (a); the only competitor of 0 (gain 0) is the post-fold radius, present iff $\lambda > \lambda_{\min}$, unique, with gain $F(\kappa_+(\lambda))/2$, whose sign is that of $\kappa_+(\lambda) - \kappa_c$ by (b); κ_+ is the inverse of the strictly increasing $\lambda|_{(\kappa_f, \infty)}$. (d): $\lambda_{\min} < \lambda_c$ from $\kappa_c > \kappa_f$; at λ_0 small radii have positive gain by (H6), so 0 is not a global maximizer and (c) forces $\lambda_c < \lambda_0$. (e): the first claim is (c), (H1) and the monotonicity of κ_+ ; for fixed b , $\lambda \mapsto G_\lambda(b)$ is convex, so $\tilde{c}$ is convex, and Danskin's theorem gives $\partial\tilde{c}(\lambda) = \text{conv}\{2bK'(2\lambda b) - R^2b^2 : b \text{ active}\}$, which at an active $b > 0$ equals $-R^2(1 - b^2)$ (using $K'(\kappa) = R^2b$) and at $b = 0$ equals $-R^2$.

(f) Fix $D \in (0, R^2]$. The function $\varphi(\lambda) := \lambda(R^2 - D) - \tilde{c}(\lambda)$ is concave with $\partial\varphi(\lambda) = \{D_b - D : b \text{ active at } \lambda\}$ convexified, where $D_b := R^2(1 - b^2)$ ($D_0 = R^2$); it is maximized at λ^* iff $D \in \text{conv}\{D_b : b \text{ active at } \lambda^*\}$. If $D_c \leq D \leq R^2$, take $\lambda^* = \lambda_c$: by (c) the active set is $\{0, b_c\}$, so $[D_c, R^2] \ni D$, and $\tilde{c}(\lambda_c) = 0$, giving $\mathcal{R}(D) = \lambda_c(R^2 - D)$. If $0 < D < D_c$, by (e) there is a unique $\lambda^* > \lambda_c$ with $D(\lambda^*) = D$; its active radius is $b = b(\kappa)$, $\kappa = \kappa_+(\lambda^*)$, with $\tilde{c}(\lambda^*) = F(\kappa)/2 = K(\kappa) - \kappa K'(\kappa)/2$ by (H4), so $\mathcal{R}(D) = \lambda^* R^2 b^2 - K(\kappa) + \kappa K'(\kappa)/2 = \kappa K'(\kappa) - K(\kappa)$ (using $\lambda^* b = \kappa/2$ and $R^2 b = K'(\kappa)$); and $b(\kappa) = \sqrt{1 - D/R^2}$ has a unique solution $\kappa > \kappa_c$ by (H1). As $D \downarrow 0$, $\kappa \rightarrow \infty$ and $\kappa K' - K = \kappa m - \log M \sim (c - a) \log \kappa \rightarrow \infty$ by Lemma 3.2. The two forms of R_c agree because $F(\kappa_c) = 0$.

(II) By Theorem 4.1(B), $\lambda(\kappa) > \lambda_0$ for all $\kappa > 0$ and λ is a bijection $(0, \infty) \rightarrow (\lambda_0, \infty)$. For $\lambda \leq \lambda_0$, G'_λ has no zero in $(0, 1]$ and $G'_\lambda(1) < 0$ (H2), so G_λ is strictly decreasing and 0 is the unique maximizer. For $\lambda > \lambda_0$ there is exactly one stationary $b_+ \in (0, 1)$; $G''_\lambda(0) > 0$ (H6) and $G'_\lambda(1) < 0$ give $G'_\lambda > 0$ on $(0, b_+)$ and < 0 on $(b_+, 1]$; $b_+(\lambda) = b(\lambda^{-1}(\lambda))$ is continuous, increasing, tends to 0 as $\lambda \downarrow \lambda_0$ and to 1 as $\lambda \rightarrow \infty$. Then $-\partial\tilde{c}(\lambda) = \{R^2\}$ for $\lambda \leq \lambda_0$ and $\{R^2(1 - b_+^2)\}$ for $\lambda > \lambda_0$, and the computation of (f) applies for every $0 < D < R^2$. $\square$

Corollary 6.2 ($\mathbb{CP}^{d-1}$). For $(a, c) = (1, d)$, Theorem 6.1 is 68BH Theorem 5.1(a)–(e) and Corollary 5.2; with 68BH Theorem A (the 1D2Q envelope, resting on the Brazitikos–Pandis extremum), $\mathcal{R} = R_d$ is the two-piece rate–distortion function of 68BH Theorem 5.1(f). $\square$

Remark 6.3 (what is proved, what must be imported). Theorem 6.1 is a statement about the one-parameter problem $\max_b \{K(2\lambda b) - \lambda R^2 b^2\}$ and its Legendre-type envelope, for which several first-order transitions are not excluded a priori [EE]. To read $\mathcal{R}$ as the rate–distortion function of a source with overlap law $\text{Beta}(a, c - a)$ one needs, as in 68BH Section 2: (i) the fixed-norm spectral extremum, that the tilted normalizer at fixed report radius is maximized by the source-orbit direction (for $\mathbb{CP}^{d-1}$: Brazitikos–Pandis through 1D2Q Proposition A.1; for $\mathbb{RP}^{d-1}$, $\mathbb{HP}^{d-1}$ the analogous Dirichlet($\beta/2, \dots, \beta/2$) statement is not in the corpus and is open); (ii) radially of the tilted (posterior) mean at the maximizer (automatic for one-spike spectra); (iii) the Gibbs-dual envelope formula and attainment of active radii by covariant channels (1D2Q Theorem 2.1, 6XH6 eq. (4.3)). Given (i)–(iii), Theorem 6.1 yields the complete two-piece rate–distortion function, exactly as in 68BH Theorem 5.1(f). Nothing in Sections 3–5 depends on these imports.

7. Bessel family and real/quaternionic projective sources

Let $Z_c(\kappa) := {}_0F_1(; c; \kappa^2/4) = \Gamma(c)(\kappa/2)^{1-c} I_{c-1}(\kappa)$, $c > 0$ (DLMF 10.25.2). For $c > \frac{1}{2}$, $Z_c(\kappa) = \mathbb{E} e^{\kappa Y}$ with $Y = 2X - 1$, $X \sim \text{Beta}(c - \frac{1}{2}, c - \frac{1}{2})$; for $c = \frac{1}{2}$, $Z_{1/2} = \cosh \kappa$ ($Y = \pm 1$); for $c = n/2$, Y is a coordinate of the uniform law on S^{n-1} ; for $c = n$, Z_n is the normalizer of the phase-lifted $\mathbb{CP}^{n-1}$ coherent orbit (6XH6 Proposition 7.1 with $k = 1$). Here $\mu = 0$, $R^2 = 1$, $K = \log Z_c$, $m = Z'_c/Z_c$, $H_Z = m - \kappa m'$. From $zw'' + cw' - w = 0$ for $w = {}_0F_1(; c; z)$ with $z = \kappa^2/4$:

$$\begin{aligned} \kappa Z''_c + (2c - 1)Z'_c - \kappa Z_c &= 0, & \kappa m' &= \kappa(1 - m^2) - (2c - 1)m, \\ H_Z &= \kappa(m^2 - 1) + 2cm = (1 - m^2)(\Psi_0(m) - \kappa), \end{aligned} \quad (7.1)$$

with $\Psi_0(m) = 2cm/(1 - m^2)$ on $(0, 1)$, and, on the curve $\kappa = \Psi_0(m)$, the denominator $\kappa(1 - m^2) - (2c - 1)m$ of $d\kappa/dm$ equals m , so

$$\Delta_0(m) := \Psi'_0(m) - \frac{\Psi_0(m)}{m} = \frac{4cm^2}{(1 - m^2)^2} > 0. \quad (7.2)$$

Also $H_Z(\kappa) = \kappa^3/(4c^2(c + 1)) + O(\kappa^5) > 0$ near 0^+ (manuscript_checks.py).

Theorem 7.1 (no fold, $c \geq \frac{1}{2}$). For $c \geq \frac{1}{2}$, $H_Z > 0$ on $(0, \infty)$; the radial multiplier $\kappa/(2m(\kappa))$ increases strictly from $\lambda_0 = c$ to ∞ , and for $c > \frac{1}{2}$ the complete radial phase and envelope are those of Theorem 6.1(II) with (a, c) replaced by $(c - \frac{1}{2}, 2c - 1)$ and κ by 2κ : continuous onset, no coexistence.

Proof. For $c \geq \frac{1}{2}$, $m = \mathbb{E}_\kappa Y \in (0, 1)$ and $m' = \text{Var}_\kappa Y > 0$, so $\kappa(m)$ exists on $(0, 1)$ and $D_0(m) := \Psi_0(m) - \kappa(m)$ has $D_0 > 0$ near 0^+ and $D'_0 = \Delta_0 > 0$ at any zero; a first zero would need $D'_0 \leq 0$: there is none, so $H_Z > 0$. The multiplier statement follows as in Theorem 4.1 ($\lambda_0 = 1/(2\text{Var}Y) = c$). For $c > \frac{1}{2}$, Kummer’s second theorem (DLMF 13.6.9 with $\nu = c - 1$) gives $Z_c(\kappa) = e^{-\kappa} {}_1F_1(c - \frac{1}{2}, 2c - 1; 2\kappa)$, i.e. $\log Z_c(\kappa) = K_{a', 2a'}(2\kappa)$ with $a' = c - \frac{1}{2}$ and $\mu' = \frac{1}{2}$, whence $H_Z(\kappa) = 2H_{a', 2a'}(2\kappa)$ and $b_Z(\kappa) = b_{a', 2a'}(2\kappa)$; this is the case $c' = 2a'$ of Theorems 4.1(B) and 6.1(II). $\square$

Proposition 7.2 ($0 < c < \frac{1}{2}$). For $0 < c < \frac{1}{2}$, $H_Z > 0$ on $(0, \infty)$ still holds, but Z_c is not a moment generating function: $m(\kappa)$ exceeds 1 for large κ ($m = 1 + (1 - 2c)/(2\kappa) + O(\kappa^{-2})$; numerically $\max m = 1.70$ for $c = 0.1$), so “onset” and “radius” have no probabilistic meaning there.

Proof. Z_c and Z'_c have positive coefficients, so $m > 0$. On the set $\{m < 1\}$, (7.1) gives $\kappa m' = \kappa(1 - m^2) + (1 - 2c)m > 0$; at a point where $m = 1$, $\kappa m' = 1 - 2c > 0$, so m cannot cross 1 downward and $\{m < 1\}$ is an initial interval $(0, \kappa_1)$ ($\kappa_1 \leq \infty$). On it the argument of Theorem 7.1 applies verbatim (only $m' > 0$, $m \in (0, 1)$, $D_0 > 0$ near 0^+ and (7.2) were used), so $H_Z > 0$ there; on $[\kappa_1, \infty)$, $m \geq 1$ and $H_Z = \kappa(m^2 - 1) + 2cm > 0$ trivially. The expansion of m is DLMF 10.40.1. $\square$

The restriction of Theorem 7.1 to $c \geq \frac{1}{2}$ and the repair in Proposition 7.2 were requested by the adversarial review; all applications below have $c \geq 1$.

Corollary 7.3. (i) *Spheres.* For $S^{n-1} \subset \mathbb{R}^n$, $n \geq 2$ ($c = n/2$), the scalar radial problem has no fold: continuous onset at $\lambda_0 = n/2$. Since $\mathbb{E}e^{\langle \kappa u, x \rangle}$ depends on u only through $|u|$, the fixed-norm extremum (i) of Remark 6.3 is automatic, and the smooth curve is consistent with the Dytso–Cardone rate–distortion function [DC]; $n = 3$ is 68BH Corollary 5.2. (ii) *Phase-lifted $\mathbb{CP}^{n-1}$ coherent states* (6XH6, $k = 1$, $c = n$): the radial problem $\phi_s(b) = K(2sb) - sb^2$ of 6XH6 eq. (5.1) is the scalar problem with $(\mu, R^2) = (0, 1)$, so 6XH6 Theorem 7.3(1) (negative fourth cumulant, continuous first activation) sharpens to the complete phase: no coexistence at any field. This source is the sphere S^{2n-1} of (i). (iii) *Oriented two-planes in $\mathbb{R}^3$, $\mathbb{R}^4$* (7H9F, $q = 1, 2$): $L_1 = \sinh \kappa / \kappa = Z_{3/2}(\kappa)$ and $L_2 = Z_{3/2}(\kappa/2)^2$, so $H_{L_2}(\kappa) = H_{Z_{3/2}}(\kappa/2)$, and Theorem 7.1 gives a second proof of 7H9F Lemma 7.1 for $n = 3, 4$ (not for $n = 5$). (iv) *The symmetric Beta case.* $c = 2a$ is the Bessel family with $c' = a + \frac{1}{2}$; in particular $\mathbb{RP}^1$, $\mathbb{CP}^1$, $\mathbb{HP}^1$ are S^1 , S^2 , S^4 .

Theorem 7.4 (rank-one sources over $\mathbb{R}, \mathbb{C}, \mathbb{H}$). Let $\beta \in \{1, 2, 4\}$, $d \geq 2$, and $(a, c) = (\beta/2, \beta d/2)$, the overlap law $|\langle x, u \rangle|^2$ of Haar unit vectors in $\mathbb{F}^d$. Then $\mu = 1/d$, $R^2 = 1 - 1/d$, $\lambda_0 = d(\beta d + 2)/4$, and the scalar radial problem has exactly one fold iff $d \geq 3$ (for all three fields), with $\kappa_f \geq (\beta d + 2)(d - 2)/d$ and $m_f \geq \frac{1}{2}$; for $\beta \in \{2, 4\}$ also $\kappa_f < \beta d(\beta + 2)/4$ ($2d$ and $6d$); for $d = 2$ the onset is continuous. Theorem 6.1 gives the complete radial phase in each case.

Proof. $|\langle x, u \rangle|^2$ is the ratio of a sum of β squared real Gaussian coordinates to the sum of βd , hence Beta($\beta/2, \beta(d - 1)/2$); $c - 2a = \beta(d - 2)/2$; apply Theorems 4.1, 5.1 ($a = \beta/2 \geq 1$ iff $\beta \geq 2$) and 6.1. $\square$

For $\beta = 2$ this is 68BH; for $\beta = 1$ (real Bingham, [B]) and $\beta = 4$ the radial statement is new, and the rate–distortion reading awaits the extremum (i) of Remark 6.3.

8. The oriented-two-plane normalizer

Let $q \geq 1$, $E(\kappa) = M_{q+1}(\kappa) = {}_1F_1(1; q + 1; \kappa)$, $O(\kappa) = E(-\kappa)$, $L_q = (E + O)/2$, $S = (E - O)/2$. Then $L_q(\kappa) = q! \sum_j \kappa^{2j} / (q + 2j)! = {}_1F_2(1; \frac{q+1}{2}, \frac{q+2}{2}; \kappa^2/4) = \mathbb{E} \cosh(\kappa T)$, $T \sim \text{Beta}(1, q)$, the normalizer of Haar oriented two-planes in $\mathbb{R}^{q+2}$ (7H9F eqs. (11), (24)); $K_q = \log L_q$, $m_q = K'_q$, $H_q = m_q - \kappa m'_q$, $J_q = L_q^2 H_q$ (7H9F eq. (25)), and 7H9F's multiplier is $\lambda_q = \kappa / (2m_q)$, i.e. $\mu = 0$, $R^2 = 1$.

Proposition 8.1 (exact structure). (i) $\kappa L'_q = \kappa S - q L_q + q$ and $\kappa S' = \kappa L_q - q S$ (68BH Lemma 3.1 at $\pm \kappa$). (ii) L_q satisfies the inhomogeneous second-order equation $\kappa^2 L'' + 2q \kappa L' - (\kappa^2 - q(q - 1))L - q(q - 1) = 0$ (homogeneous only for $q = 1$) and the homogeneous third-order equation

$$\kappa^2 L''' + (2q + 2)\kappa L'' + (q(q + 1) - \kappa^2)L' - 2\kappa L = 0. \quad (8.1)$$

(iii) In the variables $(s, u) = (S/L, 1/L)$, $u^2 \kappa J_q = -\kappa^2(1 - s^2) + \kappa s(1 + 2qu) - q(1 - u)(qu + 2)$ with $\kappa s' = \kappa(1 - s^2) - qsu$, $\kappa u' = -u(\kappa s + qu - q)$: the sign of H_q depends on the three-dimensional state (s, u, κ) , equivalently (m, m', κ) , and no first-order Riccati equation for m_q exists. (iv) Along the

trajectory,

$$H'_q + \left(3m_q + \frac{2q+2}{\kappa}\right)H_q = \Delta_q(m_q, \kappa), \quad (8.2)$$

$$\Delta_q(m, \kappa) := -\kappa m(1 - m^2) + (2q+5)m^2 - 2 + \frac{(q+1)(q+2)m}{\kappa},$$

equivalently $(\kappa^{2q+2}L_q^3H_q)' = \kappa^{2q+2}L_q^3\Delta_q(m_q(\kappa), \kappa)$. Hence, if $\kappa \mapsto \Delta_q(m_q(\kappa), \kappa)$ is negative on $(0, \kappa_\Delta)$ and positive on (κ_Δ, ∞) , then H_q has exactly one positive zero, located after κ_Δ .

Proof. (i) is linear algebra on $\kappa E' = (\kappa - q)E + q$ and $\kappa O' = q - (\kappa + q)O$; eliminating S gives (ii), and differentiating removes the constant. (iii) is substitution. (iv): with $L''/L = m' + m^2$, $L'''/L = m'' + 3mm' + m^3$, (8.1) expresses m'' through (m, m', κ) ; inserting it into $H'_q = -\kappa m''$ gives (8.2), and $(\kappa^{2q+2}L_q^3)' / (\kappa^{2q+2}L_q^3) = 3m + (2q+2)/\kappa$. All identities are checked with symbolic q in `sym_evenpart.py`, `sym_evenpart_ode.py`, `manuscript_checks.py`. For the last claim, $W := \kappa^{2q+2}L_q^3H_q$ has $W(0^+) = 0$ ($H_q = O(\kappa^3)$), decreases then increases, and $W \rightarrow +\infty$ since $H_q \rightarrow 1$. $\square$

Proposition 8.2 (exact small-field signs; found and verified in review). Along the trajectory, $\Delta_q(m_q(\kappa), \kappa)$ and H_q are even and odd in κ respectively, with

$$[\kappa^0]\Delta_q = 0, \quad [\kappa^2]\Delta_q = -\frac{4(2q+5)(q^2 - q - 8)}{(q+1)^2(q+2)^2(q+3)(q+4)}, \quad (8.3)$$

$$[\kappa^1]H_q = 0, \quad [\kappa^3]H_q = -\frac{4(q^2 - q - 8)}{(q+1)^2(q+2)^2(q+3)(q+4)}.$$

Since $q^2 - q - 8 < 0$ for $q \leq 3$ and > 0 for $q \geq 4$, both Δ_q and H_q are positive near 0^+ for $q \leq 3$ and negative for $q \geq 4$.

Proof. Series substitution in (8.2) (`manuscript_checks.py`, `reviewer/rev_last.py`). $\square$

Numerical evidence (`scan_evenpart.py`, `checks_bounds.py`; independently reproduced in review with a 4500-point grid). For $q = 1, 2, 3$: $\Delta_q > 0$ on the whole grid and H_q has no zero (consistent with 7H9F Theorem 7.2). For $4 \leq q \leq 60$: Δ_q changes sign exactly once, at κ_Δ slightly below the fold; H_q and F_q have exactly one zero each. Values: $q = 4$: $\kappa_f = 4.415714051$, $\kappa_c = 5.541056361$ (the values printed in 61Y0 Section 6, whose normalizer $12(2 \cosh \kappa - 2 - \kappa^2)/\kappa^4$ equals L_4 ; checked), $\kappa_\Delta = 4.1619$; $q = 5$: $\kappa_f = 7.61354$, $\kappa_\Delta = 7.2901$; $q = 10$: $\kappa_f = 19.37353$, $\kappa_\Delta = 18.9911$; $q = 30$: $\kappa_f = 59.99529$, $\kappa_\Delta = 59.5975$; $q = 60$: $\kappa_f = 119.9999987$. The identity (8.2) holds at the located folds to $6 \cdot 10^{-26}$.

What remains open. The single sign change of Δ_q along the trajectory is numerical. At the fold the $O(q)$ terms of Δ_q cancel and $\Delta_q(m_f, \kappa_f) = O(1)$, so no crude bound on $m_q(\kappa)$ certifies the sign change; a proof along this route needs m_q to $O(1/q)$ accuracy near $\kappa \approx 2q$. The two-dimensional transversality argument of Section 4 does not transfer because of Proposition 8.1(iii). We therefore claim no new proof of 7H9F Theorem 8.2 or Theorem 9.1 ($n \geq 6$); what the present method recovers of 7H9F is $n = 3, 4$ (Corollary 7.3(iii)). A further numerical observation: the tilted variance $V_q = K_q''$ is unimodal on the grid for $q \geq 4$ and decreasing for $q = 3$; since $H_q = \kappa(\bar{V}_q - V_q)$ with $\bar{V}_q(\kappa) = \kappa^{-1} \int_0^\kappa V_q$ the running mean, unimodality of V is a sufficient condition for one fold (and monotone decrease for none), unproved for L_q . Not covered by any result here: 7H9F for $q \geq 3$, Slater orbits with $k \geq 2$ ($_{k-1}F_k$), coherent orbits with $p \geq 2$ ($_pF_{p+1}$, 6XH6/6DJ3) and Grassmann rank $r \geq 2$ (matrix-argument $_1F_1$, 68BH open question 1); there the coefficient methods of 61Y0 Lemma 6.1 and 7H9F Theorem 8.2 remain the only proofs.

9. Numerical verification

`scan_ac.py` (mpmath, 30 digits) counts the positive zeros of H and F on a geometric-plus-linear κ -grid up to $15c + 50$ with bisection to 10^{-25} , for $a \in \{\frac{1}{2}, 1, \frac{3}{2}, 2, 3, 5\}$, $c \in (a, a + 60]$ in steps of $\frac{1}{2}$ plus the boundary values $c = 2a - \frac{1}{2}, 2a - \frac{1}{10}, 2a, 2a + 10^{-3}, 2a + 10^{-2}, 2a + \frac{1}{10}$ (744 pairs): the counts are 1, 1 for every $c > 2a$ and 0, 0 for every $c \leq 2a$, never more; $\kappa_f < \kappa_c$, $m_f \geq \frac{1}{2}$ and $\kappa_f \geq 2(c+1)(c-2a)/c$ hold in every fold case; m computed through the Kummer derivatives ${}_1F_1(a+1; c+1; \cdot)$, ${}_1F_1(a+2; c+2; \cdot)$ and through the Riccati equation (3.1) agree to $\leq 10^{-24}$ relative. The adversarial review reproduced this on an independent grid ($a \in \{0.3, 0.7, 1, 2.5, 4\}$, $c - 2a \in \{-1, -0.1, 0.1, 1, 10, 40\}$) with the same outcome. `tables_ac.py` (40 digits, 200-step bisection, Kummer-derivative route only) produces the tables below and asserts every proved relation of Theorems 4.1, 5.1 and 6.1 in each row.

Table 1: the fold and the bounds of Theorems 4.1 and 5.1 ($\kappa_1 = 2(c+1)(c-2a)/c$; the columns $c(a+1)$ and m_u apply for $a \geq 1$ only; $b_f = (m_f - \mu)/(1 - \mu)$).

(a, c)	source	κ_1	κ_f	$c(a+1)$	m_f	m_u	$\lambda_{\min}$
(1/2, 3/2)	$\mathbb{RP}^2$	1.66667	2.178287975	—	0.549065	—	3.365743198
(1/2, 2)		3	3.879009872	—	0.567380	—	4.583233767
(1/2, 5)	$\mathbb{RP}^9$	9.6	11.62828791	—	0.573922	—	11.04133789
(1, 3)	$\mathbb{CP}^2$	2.66667	3.232708836	6	0.534420	0.666667	5.358728277
(1, 5)	$\mathbb{CP}^4$	7.2	8.431550876	10	0.548068	0.6	9.689551191
(1, 17)	$\mathbb{CP}^{16}$	31.7647	33.90710030	34	0.528252	0.529412	33.99087521
(2, 6)	$\mathbb{HP}^2$	4.66667	5.277112966	18	0.521455	0.777778	9.350549631
(3, 9)		6.66667	7.295434543	36	0.515543	0.833333	13.34623024
(5, 65)		111.692	113.4549685	390	0.506602	0.846154	121.8673998
(1, 2.001)		0.0029995	0.0037492	4.002	0.500062	0.749875	3.002499797

Table 2: the contact ($D_c/R^2 = 1 - b_c^2$; $R_c = \lambda_c R^2 b_c^2$).

(a, c)	κ_c	b_c	λ_c	λ_0	D_c/R^2	R_c
(1/2, 3/2)	2.922626590	0.429029	3.406094244	3.75	0.815934	0.417964
(1/2, 2)	5.258427922	0.557080	4.719631691	6	0.689662	1.098513
(1/2, 5)	16.75539491	0.686037	12.21173305	30	0.529354	5.172667
(1, 3)	4.344285303	0.400332	5.425859459	6	0.839735	0.579718
(1, 5)	11.64408805	0.573491	10.15193548	15	0.671108	2.671112
(1, 17)	52.47145152	0.676014	38.80942424	153	0.543005	16.69245
(2, 6)	7.106601272	0.375200	9.470425005	10.5	0.859225	0.888798
(3, 9)	9.837403413	0.363856	13.51826860	15	0.867609	1.193132

Table 3: no fold for $c \leq 2a$ (the minimum of H is taken on the grid, which starts at $\kappa = 10^{-3}$ where $H \sim \kappa^3$).

(a, c)	source	zeros of H	zeros of F	$\min H$	λ_0
(1, 2)	S^2	0	0	$2.8 \cdot 10^{-12}$	3

(a, c)	source	zeros of H	zeros of F	$\min H$	λ_0
$(2, 4)$	S^4	0	0	$7.1 \cdot 10^{-13}$	5
$(1/2, 1)$	S^1	0	0	$7.8 \cdot 10^{-12}$	2
$(3/10, 3/5)$		0	0	$1.4 \cdot 10^{-11}$	1.6

Table 4: the upper bound $\kappa_f < c(a + 1)$ of Theorem 5.1 for $a < 1$, where it is not proved: violated in three of the four cases.

(a, c)	κ_f	$c(a + 1)$	ratio
$(1/4, 1)$	2.8008995	1.25	2.2407
$(1/2, 2)$	3.8790099	3	1.2930
$(1/2, 10)$	22.328044	15	1.4885
$(3/4, 3)$	4.9219982	5.25	0.9375

The $(1, 3)$ row reproduces 68BH Tables 1–3 ($\kappa_f = 3.2327088363$, $\kappa_c = 4.3442853031$, $\lambda_c = 5.42585945949$); the $q = 4$ even-part values of Section 8 reproduce 61Y0.

10. Open questions

1. A proof that $\Delta_q(m_q(\kappa), \kappa)$ changes sign once (Section 8), or a three-dimensional transversality argument in (s, u, κ) for the quadratic-in- κ sign function of Proposition 8.1(iii); either would give a second proof of 7H9F Theorem 8.2 for $q \geq 4$.
2. An upper bound on κ_f for $a < 1$ (real projective spaces), where (5.2) is unavailable.
3. The Dirichlet($\beta/2, \dots, \beta/2$) fixed-norm extremum (real and quaternionic analogue of Brazitikos–Pandis) needed to turn Theorem 7.4 into rate–distortion functions for $\mathbb{RP}^{d-1}$, $\mathbb{HP}^{d-1}$.
4. Whether the tilted variance $\text{Var}_\kappa X$ is unimodal in κ for every ${}_1F_1(a; c)$ with $c > 2a$ (an alternative proof of Theorem 4.1 and a natural conjecture for ${}_pF_{p+1}$ normalizers).
5. Whether “one fold iff right-skewed” persists for matrix-argument ${}_1F_1$ (68BH open question 1).

11. Reproducibility

All scripts are in repro/ next to this file (Python 3.12.6, sympy 1.14.0, mpmath 1.3.0; python run_all.py runs everything, writes <name>.out and runtimes.txt, and fails on any assertion; --fast skips the four slow scripts). The six scripts of the working directory were copied unchanged; the four reviewer scripts are in repro/reviewer/; manuscript_checks.py and tables_ac.py were written for this manuscript.

script	certifies	arithmetic	time
sym_riccati.py	(3.1)–(3.2), identities (a)–(d), $\Psi(\frac{1}{2})$, small- κ coefficients (3.3), the ${}_0F_1$ reduction (7.1)–(7.2), the (L, S) system	sympy exact	33 s
sym_evenpart.py	Proposition 8.1(i),(iii); small- κ order of κJ_q	sympy exact	2 s
sym_evenpart_ode.py	Proposition 8.1(ii),(iv) with symbolic q	sympy exact	2 s

script	certifies	arithmetic	time
manuscript_checks.py	every displayed identity of Sections 2–8 (48 checks), including (5.2)’s integrand, Kummer’s second theorem to $O(\kappa^{10})$, (8.3), L_4 = the 61Y0 normalizer	sympy exact	153 s
scan_evenpart.py	zero counts of H_q, F_q for $q \leq 20, 25, 30, 40, 50, 60$; (8.2) at the folds	mpmath 30 digits	4 s
checks_bounds.py	$\kappa(1 - m) < c - a$ for $a \geq 1$ and its failure for $a < 1$; (5.2) numerically; sign changes of Δ_q (delta_signchanges.json); unimodality of V_q	mpmath 30 digits	24 s
scan_ac.py	the 744-pair scan of Section 9 (scan_ac.json, scan_ac.out)	mpmath 30 digits	146 s
tables_ac.py	Tables 1–4 and the asserted relations	mpmath 40 digits	160 s
reviewer/rev_*.py (four scripts)	the reviewer’s independent re-derivations: symbolic identities (rev_sym), independent (a, c) grid (rev_num), L_q ODEs and sign changes (rev_even), (8.3) and the ${}_0F_1$ small- c behaviour (rev_last)	sympy; mpmath 30 digits	34, 90, 30, 99 s

No proof step depends on floating point: Theorems 3.1, 4.1, 5.1, 6.1, 7.1, 7.4 and Propositions 7.2, 8.1, 8.2 use only the rational identities listed, each a short hand computation machine-checked for safety, and the cited DLMF expansions.

AI-assistance statement

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